

CAN JOINT-EMBEDDING PREDICTION SUPPORT PLANNING IN LANGUAGE?

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ABSTRACT

Autoregressive language models select actions by predicting text, although planning only requires a model to preserve the consequences that determine which action should be taken. We study whether an action-conditioned joint-embedding predictive architecture can provide this narrower computational substrate. Our controlled setting presents a reasoning state and a common menu of executable intent phrases. A predictor imagines the latent successor of each phrase, and geometric advantage ranking (GAR) teaches local orderings among factual and counterfactual successors. Across five seeds on stylized iGSM, an MLP-predictor JEPA reaches $43.3 \pm 1.9\%$ strict success. This is competitive with a token language model at $42.2 \pm 12.4\%$, but below a sentence language model at $47.1 \pm 4.6\%$ and a sentence model with latent prediction at $52.1 \pm 4.7\%$. The central result is mechanistic. Latent prediction alone reaches 18.3%, removing GAR reaches 7.6%, and ranking without direct advantage regression reaches 35.8%. Local action ordering is substantially stronger through the learned transition-value score than through raw goal distance. A direct action ranker matches the full model in one-seed evaluation, which limits the present claim about predictive factorization. Repaired test-time scaling further reveals compounding rollout error rather than improvement. These results establish that non-generative latent consequence prediction can support language-action planning, while showing that useful planning geometry does not follow automatically from next-state prediction. We release a falsifiable analysis protocol for separating world-model quality, local action ordering, representation content, and test-time simulation.

1 INTRODUCTION

Language reasoning is usually studied through generation. A model emits the next token or sentence, and the resulting text becomes both its computation and its observable answer. This design is powerful, but it binds reasoning to reconstruction. An agent that chooses among language actions needs a more specific capability. It must predict how each action changes the problem, retain the aspects of that change that affect future choices, and rank the resulting alternatives.

Joint-embedding predictive architectures offer a direct way to test this possibility (LeCun, 2022). A JEPA predicts the representation of a target from a context instead of reconstructing every target token (Assran et al., 2023). Feature prediction has also been extended to video (Bardes et al., 2024). Action-conditioned variants can serve as latent world models and can support planning in physical environments (Assran et al., 2025). Yet accurate latent prediction does not guarantee that distances or scores in the learned space order actions by their usefulness. A representation can preserve the next observation while arranging successor states poorly for planning. This gap becomes especially clear for discrete language actions, where several phrases can be executable but only some advance the current proof or computation.

We isolate this issue in a controlled language-action setting. Each episode is a stylized iGSM arithmetic dependency problem. At every step, all learned systems receive the same prompt, observed history, and shuffled symbolic menu of currently executable intent phrases. The environment executes the selected phrase and reveals its textual consequence. This interface is candidate privileged relative to unrestricted text generation, but it is information matched across methods and exposes the planning mechanism without conflating it with action proposal or syntax validity.

Our model predicts a latent successor for every candidate action. We introduce geometric advantage ranking (GAR), which uses factual and counterfactual continuations to train local action orderings and a direct advantage regression term. The deployed score factors through the predicted successor. This construction permits targeted controls that remove latent prediction, remove ranking, detach ranking gradients from the predictive body, or bypass the successor with a direct action ranker.

The experiments answer four questions. First, can latent consequence prediction support closed-loop language planning? Second, is next-latent prediction sufficient? Third, does GAR create action-ordering structure that matters for planning? Fourth, what information do JEPA states retain relative to generative baselines? We also ask whether deeper latent simulation yields useful test-time scaling. The current answer to the last question is negative under a repaired leakage-free protocol.

Our contributions are as follows.

- We give a controlled formulation of language reasoning as action-conditioned latent prediction followed by local successor ranking.
- We show across five seeds that JEPA planning is viable and stable, although a sentence model with a hybrid generative and latent objective remains strongest in strict success.
- We identify GAR as the main source of planning performance. Next-state prediction alone is inadequate, and meaningful counterfactual labels matter more than additional counterfactual exposure by itself.
- We separate raw geometry, learned transition value, gradient routing, and direct discrimination. The present direct-ranker result prevents a stronger claim that predictive factorization is uniquely responsible for in-distribution performance.
- We provide frozen-feature and local-ordering analyses, then state the remaining controlled minimal-pair and cross-domain tests as explicit empirical obligations.

2 A FRAMEWORK FOR LATENT LANGUAGE PLANNING

2.1 LANGUAGE ACTIONS AND CONSEQUENCES

Let x denote a problem statement, h_t the observed history at step t , and $s_t = (x, h_t)$ the corresponding textual state. The environment exposes a finite menu $\mathcal{A}(s_t)$ of executable intent phrases. Executing $a_t \in \mathcal{A}(s_t)$ produces a textual observation o_{t+1} and a new state s_{t+1} . An episode succeeds when the requested quantity is resolved. Strict success requires the minimum number of actions. Slack- k success permits at most k excess actions.

The shared menu removes action proposal from the main comparison. It supplies legality at the current state, but not action values, the optimal action, future menus, or a solved state. We use symbolic and future-action information only in labeled diagnostics. Thus the primary question is conditional planning: given identical executable candidates, which training objective creates the most useful decision representation?

2.2 JOINT-EMBEDDING PREDICTION

An online encoder f_θ maps a textual state to $z_t = f_\theta(s_t) \in \mathbb{R}^d$. An action encoder g_θ maps an intent phrase to $u_t = g_\theta(a_t)$. A predictor F_θ produces

$$\hat{z}_{t+1} = F_\theta(z_t, u_t). \quad (1)$$

The target $\bar{z}_{t+1} = f_{\bar{\theta}}(s_{t+1})$ comes from a stopped-gradient exponential-moving-average encoder. The target encoder remains in evaluation mode, and all model dropout is zero. The factual prediction loss is

$$\mathcal{L}_{\text{pred}} = \|\hat{z}_{t+1} - \text{sg}(\bar{z}_{t+1})\|_1. \quad (2)$$

Exact counterfactual transitions supply the same loss for alternative executable actions. These alternatives are generated by the environment and do not use model rollouts.

This objective can learn dynamics without producing the next sentence. It does not impose an action preference. Low prediction error and non-collapsed covariance are therefore necessary diagnostics rather than sufficient planning criteria.

2.3 PLANNING GEOMETRY AS LOCAL ORDERING

Suppose two feasible actions a^+ and a^- have continuation advantages $A_N(s, a^+) > A_N(s, a^-)$. Here A_N measures the consequence of taking a and following a reference continuation for N steps. A planning representation should order their predicted successors accordingly. Let $V_\phi(\hat{z}_{t+1}, z_t)$ be a learned transition-value score. The deployed action score is

$$Q_{\theta, \phi}(s_t, a) = V_\phi(F_\theta(f_\theta(s_t), g_\theta(a)), f_\theta(s_t)). \quad (3)$$

The predictive successor is therefore an explicit decision bottleneck. We assess local planning geometry through pairwise ranking accuracy, top-1 optimal-action recall, action regret, rank correlation with exact continuation advantage, and the positive-versus-best-negative margin. These quantities make a narrower and more testable claim than global cluster quality.

2.4 WHY TEST-TIME SIMULATION MIGHT HELP

The factorization permits recursive simulation. A depth- D planner applies F_θ along candidate action sequences and selects the first action from the best sequence. If prediction error remains controlled and the value score transfers to imagined states, deeper search can improve decisions without adding parameters. If errors compound or the scorer is calibrated only on one-step successors, more computation can reduce performance. Test-time scaling is therefore an empirical property, not a consequence of using a JEPA.

3 GEOMETRIC ADVANTAGE RANKING

3.1 OBJECTIVE

GAR trains the deployed score on within-state counterfactual comparisons. For an ordered pair (a^+, a^-) , define $q^+ = Q(s, a^+)$ and $q^- = Q(s, a^-)$. We minimize

$$\mathcal{L}_{\text{rank}} = \log(1 + \exp[-(q^+ - q^-)]) . \quad (4)$$

We also regress the exact N -step advantage target $A_N(s, a)$,

$$\mathcal{L}_{\text{adv}} = (Q(s, a) - A_N(s, a))^2 . \quad (5)$$

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \text{lambda}_{\text{cf}} \mathcal{L}_{\text{cf}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} . \quad (6)$$

Unless stated otherwise, the current recipe uses horizon $N = 2$, two counterfactual alternatives, and direct-advantage weight 0.25. Gradients from both GAR terms reach the scorer, predictor, action encoder, and online state encoder. The EMA encoder follows the online encoder slowly and never receives gradients.

3.2 WHAT GAR CAN AND CANNOT ESTABLISH

GAR has two possible roles. Its gradients can reshape the predictive representation, and its score head can learn a useful supervised readout. We separate these with four controls. *Detached geometry* blocks GAR gradients into the encoder and predictor. *Geometry only* plans with raw predicted successor distance to an encoded solved state. This is an oracle-terminal diagnostic. *Direct ranker* learns $D(s, a)$ from the same positive and counterfactual actions while bypassing the predicted successor at decision time. *Counterfactual-only* adds exact alternative-transition prediction but removes GAR.

An additional shuffled-label control retains the number and form of counterfactual examples while randomizing their action-quality correspondence. A pure direct ranker without successor auxiliaries tests whether the counterfactual data can support ordinary supervised action selection. Together these controls separate additional data, meaningful ordering labels, a useful score head, and predictive factorization.

3.3 MODELS USED IN THE COMPARISON

The MLP JEPA uses a 256-dimensional state and action representation with an MLP predictor. The causal JEPA replaces this predictor with a causal Transformer over the encoded history. The token language model scores candidate phrases through token likelihood. The sentence language model predicts the next intent as a sentence-level class. The sentence-plus-latent model adds an auxiliary next-sentence representation loss and still selects by its generative score. All systems receive the same shuffled current feasible-action menu.

We also train weight-shared recurrent token and sentence models. Their number of loops is sampled during training following the recurrent-depth test-time-compute setting of Geiping et al. (2025). Current loop curves are exploratory because their measured FLOPs are not yet calibrated to fixed work across policies.

4 EXPERIMENTAL DESIGN

4.1 ENVIRONMENT AND DATA

The completed experiments use a stylized iGSM generator derived from the controlled arithmetic dependency setting of Ye et al. (2025). Problems contain named quantities, arithmetic relations, distractors, and a target query. Training examples are generated online rather than drawn repeatedly from a fixed corpus. The paper recipe sees 300,000 examples with solution lengths from three to nine actions. Validation uses 200 fresh episodes per seed for the five-seed tables. The decision audit uses 500 episodes and the repaired scaling gate uses 1,000 paired episodes.

At each step, the current feasible menu contains executable intent phrases only. It does not reveal which action is necessary. Candidate order is independently shuffled. The random, token, sentence, and JEPA policies act in closed loop and observe the same environment outcomes. Headline results do not use an oracle solved-state embedding or future feasible-action tree.

The planned final suite includes faithful iGSM, ProofWriter (Tafjord et al., 2021), PlanBench Blocksworld (Valmeekam et al., 2023), and ALFWorld (Shridhar et al., 2021). These domains are not placed in the main result table until their action interfaces pass executor, split, feasibility-recall, and leakage gates. Empty cells record this incomplete scope.

4.2 BASELINES, TUNING, AND UNCERTAINTY

Each completed headline row uses five training seeds. Hyperparameters were selected on validation data. The current draft reports the sample standard deviation across seed means. The final paper will add paired bootstrap confidence intervals on identical generated puzzles. Random and oracle policies provide environment bounds. The direct ranker receives the same histories, candidates, positive actions, and counterfactual negatives as GAR.

The preregistered final tuning protocol chooses one learning rate per model and dataset at width 256, then reuses it for model ablations unless the loss scale or gradient path changes. Such changes receive a local learning-rate cross-check. Test results never select a learning rate. Parameter counts, training FLOPs, and the complete search ledger remain pending for the final comparison.

4.3 EVALUATION

The primary metric is strict task success. We also report success with two excess actions because it separates choosing useful actions from choosing them in the shortest order. The final evaluation will include slack zero, one, two, and four, along with area under the success-versus-budget curve, invalid-action rate, and mean excess actions among solved episodes.

Length generalization holds the distractor construction fixed while increasing the number of necessary actions beyond the training range. Only sentence and sentence-plus-latent curves have completed this corrected evaluation. We treat them as preliminary.

Table 1: Closed-loop planning on stylized iGSM. Values are mean $\pm$ sample standard deviation over five seeds. Every model receives the same shuffled menu of currently feasible actions.

Model	Strict success	Success with +2 actions
Sentence LM + latent	.521 $\pm$.047	.809 $\pm$.020
Sentence LM	.471 $\pm$.046	.813 $\pm$.023
MLP JEPA + GAR	.433 $\pm$.019	.796 $\pm$.033
Token LM	.422 $\pm$.124	.836 $\pm$.073
Causal JEPA + GAR	.366 $\pm$.047	.778 $\pm$.043
Random	<i>pending</i>	<i>pending</i>

4.4 REPRESENTATION AND DECISION DIAGNOSTICS

Frozen features are sampled immediately before the next action, with no future text. Ridge and logistic probes use problem-grouped splits. Current probes measure operation identity, remaining steps, resolved-variable count, and numerical outcome. Effective rank is computed from the feature covariance. Necessary-action balanced accuracy is exactly chance under the current probe and is excluded as positive evidence.

The decision audit enumerates identical feasible actions at aligned states and includes forced-error histories. For JEPA, it compares the deployed predicted transition-value score with raw predicted geometry. Exact continuation cost is used only as an evaluation label. The planned controlled minimal-pair suite measures paraphrase invariance, negation sensitivity, operator swaps, entity renaming, and matched-step states with different true progress.

4.5 TEST-TIME-COMPUTE VALIDITY

Our first deep-planning evaluator returned terminal action sequences early and included path length in the score. It therefore exposed solved candidates through candidate length. We reject that result. The repair uses fixed-depth continuations, an absorbing no-op after solution, equal first-action candidate budgets, randomized order, and constant path-length offsets. Zero-score and shuffled-score controls must remain near their depth-one random bounds. Depths greater than one still use a reference-graph future feasible-action tree and are labeled candidate privileged.

5 PLANNING RESULTS

5.1 LATENT PREDICTION SUPPORTS COMPETITIVE PLANNING

Table 1 and Figure 1 report the completed five-seed comparison. MLP JEPA reaches 43.3% strict success and has the smallest seed variability among learned models. Its mean is close to the token LM at 42.2% and below the sentence LM at 47.1%. The sentence-plus-latent model leads strict success at 52.1%. The causal JEPA reaches 36.6%, so replacing the MLP predictor with the present causal Transformer does not improve the recipe.

These results support viability rather than superiority. They also suggest that generative and latent-predictive objectives can be complementary. The best strict result comes from a sentence model that combines them.

5.2 NEXT-STATE PREDICTION IS NOT ENOUGH

GAR is the clearest causal result. Removing GAR reduces strict success from .433 to .076. Training only factual and counterfactual latent MSE reaches .183. Ranking without direct advantage regression reaches .358. Removing counterfactual latent prediction while retaining GAR reaches .345. Thus ordering supervision and successor prediction provide complementary gains, with ordering responsible for the larger share.

The advantage horizon is not monotonic. Horizon two reaches .433, while horizons one and four reach .161 and .181. This result concerns the current reference-continuation target and does not

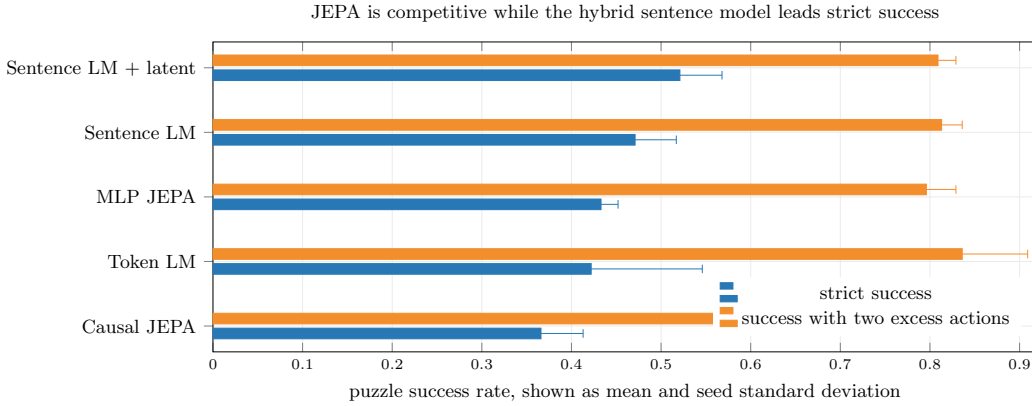


Figure 1: Five-seed planning comparison. The MLP JEPA is competitive and stable, while sentence prediction with a latent auxiliary has the strongest strict mean.

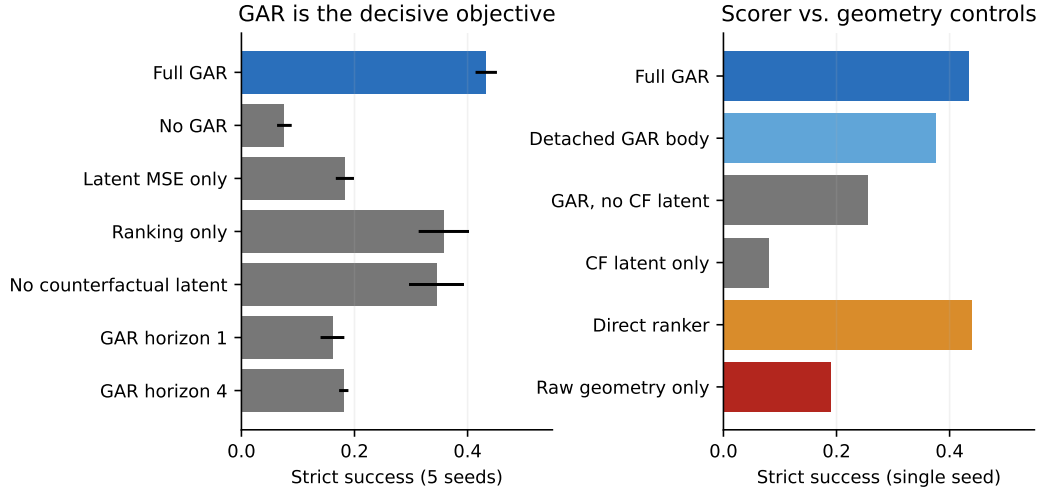


Figure 2: Left: five-seed GAR ablations. Right: one-seed mechanism controls, shown as exploratory evidence. Raw goal distance is weak, and a direct ranker matches full GAR in this in-distribution test.

establish a general optimal horizon. Increasing the number of counterfactual alternatives from one to four yields a modest change from .403 to .439. Direct advantage weights of .10 and .50 reach .443 and .402, which argues against a single fragile coefficient accounting for the full effect.

5.3 MEANINGFUL COUNTERFACTUAL LABELS MATTER

A shuffled-GAR control preserves the counterfactual examples but randomizes their quality labels. Across three learning rates it reaches .015 strict success. A direct ranker without successor auxiliaries reaches .150 to .185 across the same local learning-rate screen. By comparison, the matched full GAR seed reaches .435, and the direct ranker with JEPA successor auxiliaries reaches .440. These controls rule out the simple explanation that GAR works only because it supplies more alternative

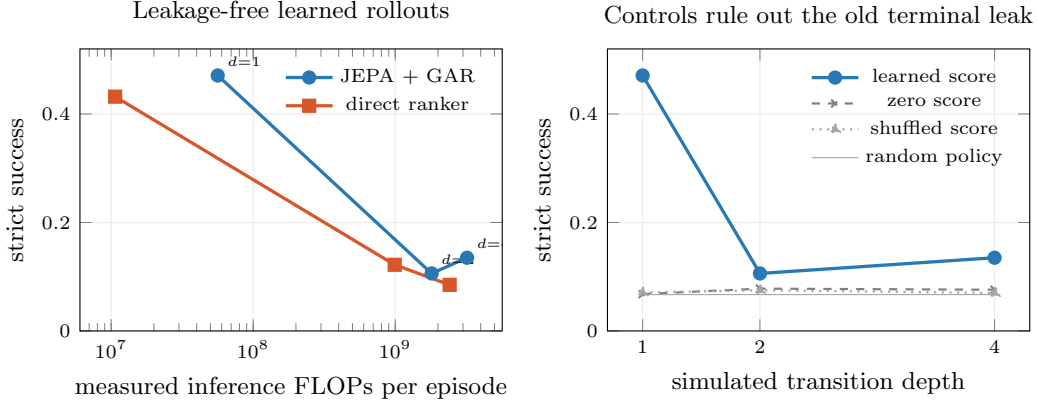


Figure 3: Repaired one-seed test-time-compute gate on 1,000 paired episodes. Depths greater than one use a candidate-privileged future feasible-action tree. Measured FLOPs increase, but learned performance falls. This curve is a diagnostic, not a final method comparison.

transitions. They also show that successor auxiliaries improve a direct scorer even when the predicted successor is not its decision bottleneck.

5.4 CURRENT PREDICTIVE FACTORIZATION CLAIM IS LIMITED

One-seed controls localize the mechanism but do not settle it. Detached GAR gradients reach .375 strict success, compared with .435 for full GAR. Geometry-only planning reaches .190. The direct ranker reaches .440. The predictive transition-value score attains .786 top-1 optimal-action recall and .826 pairwise accuracy. Raw predicted geometry reaches .549 and .603.

The learned score is therefore much more useful than Euclidean goal distance. GAR gradients into the predictive body appear helpful in one seed, but a direct discriminative path remains equally strong on in-distribution iGSM. A unique benefit of predictive factorization must be demonstrated through data efficiency, compositional length, action paraphrases, goal transfer, or test-time simulation.

5.5 DEEPER SIMULATION CURRENTLY AMPLIFIES ERROR

Figure 3 presents the repaired fixed-checkpoint gate. MLP JEPA falls from .471 at depth one to .106 at depth two and .135 at depth four. The direct ranker falls from .432 to .122 and .085. Zero and shuffled controls remain close to the random-policy rate, so the repaired evaluator removes the earlier terminal-path artifact. Additional computation does not improve the current models. The result instead identifies rollout drift or off-distribution score calibration as the next mechanism to solve.

6 WHAT THE REPRESENTATIONS ENCODE

6.1 GLOBAL RANK DOES NOT EXPLAIN PLANNING

Both JEPA variants are non-collapsed. Their effective ranks are 198.7 and 215.3 in a 256-dimensional space. The token LM has effective rank 3.5 and remains competitive in closed-loop success. Sentence and sentence-plus-latent models have ranks 79.6 and 71.4. Effective rank is therefore a representation health check, not a planning metric.

The same caution applies to global probes. The sentence-plus-latent model makes operation identity highly linearly accessible at .807 balanced accuracy. MLP JEPA reaches .406. MLP JEPA nevertheless exposes resolved-variable count with $R^2 = .895$. Remaining-step prediction reaches $R^2 = .346$ for MLP JEPA, compared with .439 for the token LM and .429 for sentence plus latent.

Numerical outcomes are weakly decoded across all systems. The current necessary-action probe has .500 balanced accuracy and provides no positive evidence.

Figure 4 in the appendix summarizes these frozen-feature diagnostics.

6.2 THE PLANNING CLAIM IS LOCAL

GAR asserts that successor differences within the same state order actions. The strongest current evidence is the gap between the deployed transition-value score and raw predicted geometry. The final analysis will report pairwise feasible-action ranking accuracy, top-1 optimal recall, mean regret, Spearman correlation with exact continuation advantage, and the positive-versus-best-negative margin at every retained checkpoint. Across seeds and checkpoints, we will compare their correlation with strict success against latent MSE, effective rank, and the global probes.

This test can falsify the proposed account. If one-step prediction error or global probe scores track planning better than local margins, then GAR should not be described as creating planning geometry. If local action ordering predicts success while global rank does not, then the representation supports planning through a low-dimensional local structure even when its full covariance retains many other factors.

6.3 CONTROLLED SEMANTIC GEOMETRY IS STILL MISSING

Ordinary t-SNE or UMAP plots cannot establish semantic invariance. The required data must hold the environment state fixed while changing wording, or hold lexical overlap fixed while changing consequence. Figure 5 in the appendix is intentionally empty in this draft. The completed version will juxtapose JEPA, token LM, sentence LM, sentence-plus-latent, no-GAR JEPA, and direct-ranker features on the same controlled pairs. Quantitative retrieval, distance, and action-ranking changes will accompany every projection.

6.4 PRELIMINARY LENGTH GENERALIZATION

The corrected distractor-matched length evaluation is complete only for the two sentence families and one seed. Both degrade beyond the training range. We do not compare methods from this partial figure. Figure 6 is placed in the appendix. The final plot requires all headline families on identical generated instances, five seeds, and confidence intervals for paired differences.

7 BACKGROUND AND RELATED WORK

Joint-embedding prediction. JEPA was proposed as a route to predictive representations that avoid reconstructing every detail of an observation (LeCun, 2022). I-JEPA predicts representations of masked image regions from visible context (Assran et al., 2023). V-JEPA extends feature prediction to video and studies motion-sensitive representations (Bardes et al., 2024). V-JEPA 2 combines large-scale action-free video pretraining with an action-conditioned model for robot planning toward image goals (Assran et al., 2025). These works motivate latent prediction, but they do not imply that metric distance ranks discrete language actions.

Language-focused JEPA work is emerging. LLM-JEPA combines latent prediction with language-model training (Huang et al., 2026). JEPA-Reasoner studies reasoning through latent predictive objectives (Liu et al., 2026). Our scope is narrower. We study controlled language actions with exact consequences and within-state counterfactual orderings. The resulting experiment separates action proposal, transition prediction, and candidate ranking.

Planning-oriented representation shaping. Several recent methods address the mismatch between prediction and planning. Value-guided JEPA shapes embedding distance to approximate a goal-conditioned reaching value (Destrade et al., 2026). Temporal straightening reduces latent-trajectory curvature so Euclidean distance better approximates planning-relevant geodesics (Wang et al., 2026). Temporal-Distance JEPA learns a directed temporal cost using trajectory order, cross-trajectory negatives, and rollout consistency (Bai & Xiong, 2026). GAR shares the goal of making latent

structure useful for action selection. It instead imposes local within-state order constraints over discrete language actions and trains an amortized score that factors through predicted successors.

Direct numerical comparison with visual continuous-control planners would not be meaningful because observations, action spaces, supervision, and evaluation differ. We do need matched conceptual controls. Raw distance, detached geometry, direct ranking, and exact transition diagnostics test whether our result is due to metric shaping, a supervised readout, or predictive dynamics. A temporal-order or straightening auxiliary on iGSM becomes necessary if we claim empirical advantage over those objectives. The present paper does not make that claim.

Latent world models and planning. Latent dynamics have long supported model-based control. Dreamer learns compact recurrent state-space models for imagined policy learning (Hafner et al., 2020). MuZero learns dynamics, reward, policy, and value representations without reconstructing observations (Schrittwieser et al., 2020). These systems demonstrate that a predictive state can be decision useful when policy and value losses shape it. GAR can be understood as an action-level ordering objective for an analogous language-action setting, with exact counterfactual transitions available for diagnosis.

Test-time computation. Autoregressive reasoning often allocates more inference computation through longer traces, sampling, search, or verification (Snell et al., 2024). Weight-shared recurrent Transformers permit depth to vary without increasing parameters. The recurrent-depth recipe of Geiping et al. (2025) samples loop counts during training and evaluates extrapolation at inference. JEPA rollouts offer another route because latent states can be simulated without decoding full text. Our repaired negative result shows that this potential depends on rollout stability and value calibration.

Controlled reasoning environments. iGSM generates arithmetic dependency graphs with controlled reasoning complexity (Ye et al., 2025). ProofWriter provides natural-language rule reasoning and explicit proofs (Tafjord et al., 2021). PlanBench translates classical planning tasks into language while retaining executable symbolic validation (Valmeekam et al., 2023). ALFWorld connects language instructions with interactive household environments (Shridhar et al., 2021). These environments offer complementary tests of arithmetic, deduction, planning, and grounded interaction. The current empirical claims are restricted to stylized iGSM until the remaining adapters pass their validity gates.

8 CONCLUSION

Action-conditioned joint-embedding prediction can support nontrivial language-action planning without reconstructing the next sentence. The result is competitive under a common feasible-action interface, but it does not surpass the strongest sentence baseline. The scientific contribution lies in the mechanism. Next-state prediction alone does not produce useful action selection. GAR supplies local counterfactual order constraints and a learned transition-value score, and both ranking and successor supervision matter.

The same controls sharpen the open question. Raw goal distance is weak, a direct ranker matches the current JEPA in one seed, and deeper latent rollout amplifies error. A persuasive account of JEPA reasoning must therefore show where predictive factorization helps, connect local ordering geometry to closed-loop behavior, and succeed under semantic and compositional interventions. The draft makes these obligations visible so that later results can strengthen or revise the claim without changing the evaluation contract.

The current claim is limited to conditional planning under a symbolic feasible-action menu on stylized iGSM. Cross-domain evidence and an unrestricted proposal interface remain future work.

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A LIMITATIONS AND SCOPE

The main interface supplies the current symbolic feasible-action menu. This isolates action selection but does not test unrestricted proposal or language generation. The results establish conditional planning among executable phrases. They do not establish an autonomous language agent.

The completed evidence comes from a stylized arithmetic environment. Templated language and exact counterfactual transitions are valuable for mechanism analysis, but they limit external validity. Faithful iGSM and at least one non-arithmetic domain are required before submission.

Several mechanism controls use one seed. Direct ranking currently matches full JEPA in that setting. Detached geometry, geometry-only planning, local metric correlations, and controlled semantic interventions need five-seed completion before they support a central claim.

The repaired deeper-planning protocol uses a future feasible-action tree beyond depth one. It is candidate privileged and still shows declining performance. A deployable test-time-scaling result requires a shared learned proposal interface or full-catalogue search, paired fixed-work evaluation, and calibrated FLOPs for recurrent baselines.

The present model is trained from scratch on a controlled language. Its findings do not directly establish how pretrained large language models would organize open-domain planning states. Conversely, the controlled setting enables exact causal comparisons that would be difficult to interpret in an open-ended agent benchmark.

B DETAILED PROTOCOL

Information boundary. Headline policies observe the problem statement, prior selected intent phrases and outcomes, and a shuffled menu of actions executable at the current state. They do not observe the expert action, exact remaining distance, future action availability, terminal state, or counterfactual outcomes at evaluation. Training uses exact alternative outcomes only for declared counterfactual prediction and GAR targets.

Target encoder. The EMA encoder is updated from the online encoder and kept in evaluation mode at all times. Gradients do not enter EMA targets. Dropout is disabled in the encoder, predictor, action encoder, scorer, and target encoder. These conditions follow earlier validity tests that found stochastic or train-mode targets destabilize dense rollout supervision.

Paired evaluation. Final comparisons will use identical generated puzzle indices, action-order seeds, and budgets across systems. We will report per-seed means, sample standard deviations, paired bootstrap intervals, and paired differences for headline claims. The currently completed five-seed results use 200 fresh validation episodes per seed. The fixed-checkpoint scaling gate uses 1,000 paired episodes.

Scaling controls. The repaired continuation generator allocates equal candidate budget to each root action up to one unavoidable rounding candidate. Every continuation has the requested depth. Solved paths enter an absorbing no-op state. Candidate order is randomized at every decision. The scoring offset is constant across candidates. A score-blind method therefore cannot infer success from path length, count, or order.

Tuning contract. For the final suite, width-256 learning rates are screened on validation and reused within a model-dataset pair. The planned grid is $\{5e-5, 7e-5, 10^{-4}, 3e-4, 5e-4, 7e-4, 10^{-3}, 3e-3, 5e-3\}$. The initial stage uses a compact subset. Five seeds are trained only after selecting the validation learning rate. Ablations reuse that rate unless they change gradient routing or loss scale.

Why current and historical numbers differ. Historical reports used different generators, predictor variants, batch sizes, candidate interfaces, and training budgets. This draft includes only the 300,000-example paper family and matched controls retrieved from its run lineage. Older values are retained in the repository but are not combined with the current table.

C COMPLETE CURRENT RESULT TABLES

Table 2: Five-seed GAR ablations on stylized iGSM.

Variant	Strict	+2 actions
Full GAR	.433 $\pm$.019	.796 $\pm$.033
No GAR	.076 $\pm$.013	.422 $\pm$.022
Latent MSE only	.183 $\pm$.016	.628 $\pm$.031
Ranking only	.358 $\pm$.045	.750 $\pm$.018
No counterfactual latent loss	.345 $\pm$.049	.734 $\pm$.017
GAR horizon 1	.161 $\pm$.021	.597 $\pm$.037
GAR horizon 2	.433 $\pm$.019	.796 $\pm$.033
GAR horizon 4	.181 $\pm$.008	.651 $\pm$.014
One counterfactual	.403 $\pm$.041	.772 $\pm$.020
Four counterfactuals	.439 $\pm$.049	.821 $\pm$.025
Advantage MSE weight .10	.443 $\pm$.037	.802 $\pm$.034
Advantage MSE weight .50	.402 $\pm$.039	.783 $\pm$.026

Table 3: One-seed mechanism controls. These rows are exploratory. Geometry-only uses an oracle-terminal embedding and is candidate privileged.

Variant	Strict	+2 actions
Full GAR	.435	.820
Detached GAR body	.375	.775
GAR without counterfactual latent loss	.255	.695
Counterfactual latent only	.080	.410
Direct ranker with successor auxiliaries	.440	.820
Raw geometry only [†]	.190	.625

Table 4: Five-seed representation diagnostics. Probe values are means across seeds.

Model	Effective rank	Operation bal. acc.	Remaining R^2	Resolved R^2	Outcome R^2
MLP JEPa	198.70	.406	.346	.895	.088
Causal JEPa	215.34	.403	.332	.846	.053
Token LM	3.55	.532	.439	.986	-.020
Sentence LM	79.64	.777	.410	.879	-.027
Sentence LM + latent	71.44	.807	.429	.886	-.029

D PREDECLARED ANALYSES AND DECISION RULES

Does GAR create planning geometry? For every feasible action pair at aligned states, report pairwise ranking accuracy, top-1 optimal recall, mean action regret, Spearman correlation with exact continuation advantage, and the positive-versus-best-negative margin. Across seeds and retained checkpoints, correlate each quantity with strict planning success. Compare those correlations with one-step latent error, successor retrieval, effective rank, and global probe scores. The planning-geometry account is supported only if local ordering and margin track success materially better.

Does GAR merely add counterfactual data? Compare full GAR, counterfactual latent prediction without ranking, shuffled GAR labels, a pure direct ranker without successor loss, a direct ranker with successor auxiliaries, and full GAR at matched counterfactual exposure. The completed shuffled-label and pure-direct pilots already reject the strongest data-volume explanation. Five-seed completion is required for the final claim.

Table 5: One-seed information controls across a local learning-rate screen. These controls separate meaningful counterfactual ordering from counterfactual exposure alone.

Control	Learning rate	Strict	+2 actions
Direct ranker without successor auxiliaries	10^{-4}	.180	.590
Direct ranker without successor auxiliaries	3×10^{-4}	.185	.635
Direct ranker without successor auxiliaries	7×10^{-4}	.150	.560
GAR with shuffled quality labels	10^{-4}	.015	.170
GAR with shuffled quality labels	3×10^{-4}	.015	.175
GAR with shuffled quality labels	7×10^{-4}	.015	.175

Table 6: Exploratory recurrent-depth language-model curves. These are raw one-seed loop results. FLOP calibration under fixed candidate work is pending.

Model	Metric	1 loop	2 loops	4 loops	8 loops	16 loops	32 loops
Token LM	Strict	.065	.140	.155	.170	.170	.140
Token LM	+2	.550	.700	.705	.725	.695	.650
Sentence LM	Strict	.290	.430	.475	.450	.395	<i>pending</i>
Sentence LM	+2	.700	.800	.800	.740	.735	<i>pending</i>
Sentence LM + latent	Strict	.315	.340	.360	.385	.355	<i>pending</i>
Sentence LM + latent	+2	.705	.770	.770	.795	.785	<i>pending</i>

Does predictive factorization help? Compare full GAR and the direct ranker on training-data fractions, held-out reasoning lengths, novel lexical templates, action paraphrases, negated minimal pairs, entity renaming, and novel goals. The predictive factorization is useful only if it improves at least one preregistered generalization or compute axis with uncertainty that excludes a negligible effect.

Why do generative models lead strict success? At identical aligned states, decompose errors into candidate ranking, transition prediction, recovery after a forced distractor, calibration by action type, and lexical-template sensitivity. Score sentence-plus-latent both by token likelihood and latent distance. If likelihood is strong while latent selection is weak, its advantage lies in generative readout. If both are strong, its sentence representation is a better state estimator.

Controlled geometry. Construct paired states for semantic paraphrase, action paraphrase, negation, operator swap, irrelevant entity renaming, and matched elapsed step with different progress. Report full-space retrieval and pairwise distances first. Report UMAP and t-SNE only as illustrations. Use identical frozen checkpoints, feature boundaries, projection hyperparameters, and point sets across models.

Test-time scaling. First diagnose recursive drift at each horizon, including transition error, nearest-successor retrieval, score calibration on imagined versus true states, and action-ranking degradation. Train with dense multi-step latent supervision only after this localization. Compare against Poisson-loop-trained token and sentence models on paired puzzles, with inference FLOPs measured under fixed candidate work. Plot every tested depth rather than only the best point.

E REPRODUCIBILITY AND ARTIFACT STATUS

Every figure in this draft is a vector PDF. The headline, GAR, probe, and length figures are generated from the retrieved current-findings ledger. The repaired scaling figure reads the leakage-free fixed-checkpoint gate from 2026-08-04. Exact repository paths and run identifiers are recorded in the paper README. The minimal-pair panel is explicitly marked pending and contains no data.

The completed runs use immutable code snapshots and checkpoint hashes. Failed and superseded jobs remain visible in the controller ledger. The manuscript excludes the earlier candidate-length-leaking scaling curves from positive evidence. The final artifact will provide a machine-readable manifest with checkpoint path, hash, model configuration, seed, validation split, evaluation indices, parameter count, optimizer updates, wall time, and measured FLOPs for every included row.

Table 7: Leakage-free one-seed scaling gate on 1,000 paired episodes. Depth greater than one uses a candidate-privileged future action tree.

Score	Depth 1	Depth 2	Depth 4
JEPA + GAR	.471	.106	.135
Direct ranker with successor auxiliaries	.432	.122	.085
Zero score	.068	.078	.076
Shuffled JEPA score	.071	.075	.070
Exact symbolic transition + direct score [†]	.432	.141	.087

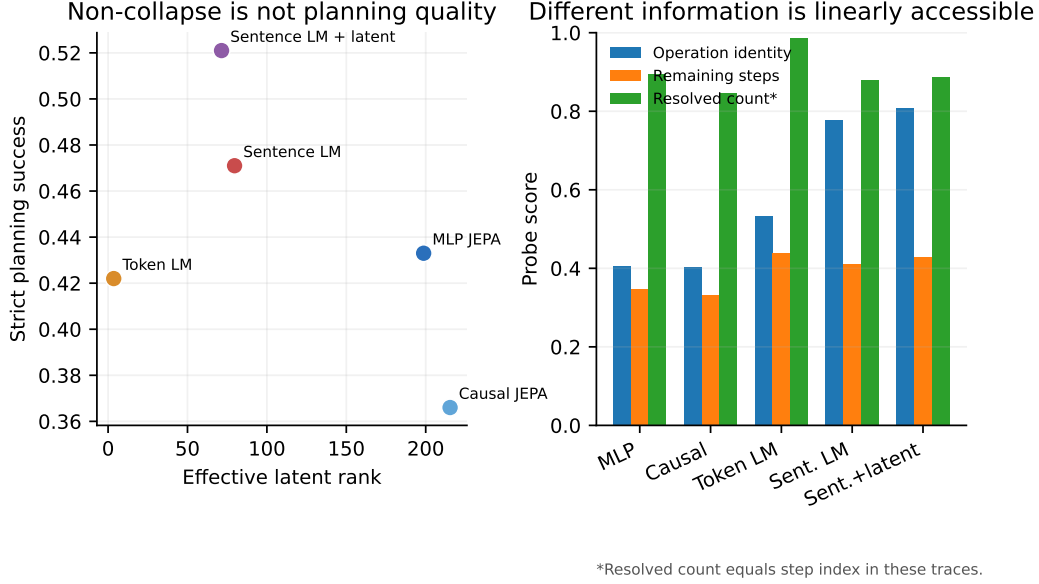


Figure 4: Five-seed frozen-feature diagnostics. Effective rank is not aligned with planning success. Sentence objectives make operation identity more accessible, while all systems expose progress-related variables. Resolved count remains confounded with elapsed step index.

The official ICLR 2027 style files are vendored unchanged in the paper directory. Run `make figures` through `make_figures.sh`, then compile `main.tex` with `latexmk -pdf`. The draft remains anonymous because `\iclrfinalcopy` is disabled.

Table 8: Paper-suite completion matrix. Empty entries are intentional.

Experiment family	Stylized iGSM	Faithful iGSM	ProofWriter	Blocksworld / ALFWorld
Five-seed headline	complete	<i>pending</i>	<i>pending</i>	<i>pending</i>
Five-seed GAR ablation	complete	<i>pending</i>	<i>pending</i>	<i>pending</i>
Local ordering audit	one seed	<i>pending</i>	<i>pending</i>	<i>pending</i>
Length OOD	partial	<i>pending</i>	<i>pending</i>	<i>pending</i>
Controlled minimal pairs	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
Test-time FLOP scaling	repaired pilot	<i>pending</i>	<i>pending</i>	<i>pending</i>

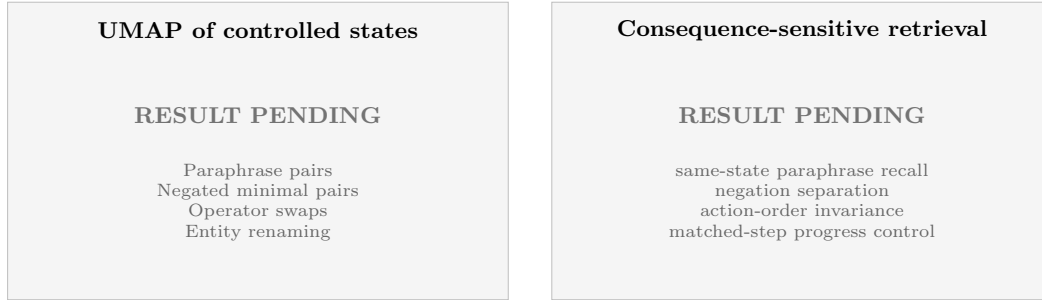


Figure 5: Reserved controlled minimal-pair analysis. No empirical points are shown because paraphrase and negation features have not yet been collected. The final visualization will be regenerated from frozen checkpoints and paired semantic interventions.

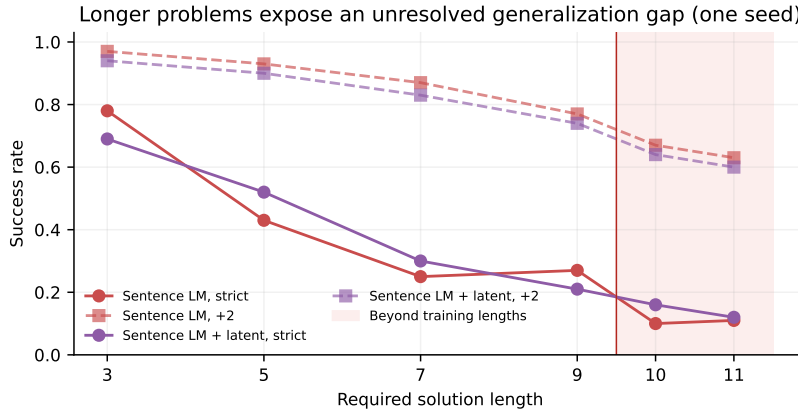


Figure 6: Preliminary length generalization for two sentence families. These one-seed curves are incomplete and do not support a cross-method conclusion.